

ROGER C. REED

The SUPERALLOYS

Fundamentals and Applications



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The Superalloys

Superalloys are unique high temperature materials used in gas turbine engines, which display excellent resistance to mechanical and chemical degradation. This book presents the underlying metallurgical principles which have guided their development and practical aspects of component design and fabrication from an engineering standpoint. The topics of alloy design, process development, component engineering, lifetime estimation and materials behaviour are described, with emphasis on critical components such as turbine blades and discs.

The first introductory text on this class of materials, it will provide a strong grounding for those studying physical metallurgy at the advanced level, as well as practising engineers. Included at the end of each chapter are exercises designed to test the reader's understanding of the underlying principles presented. Additional resources for this title are available at www.cambridge.org/9780521859042.

ROGER C. REED is Professor and Chair in Materials Science and Engineering at Imperial College London. From 1994 to 2002, he was Assistant Director of Research in the Rolls-Royce University Technology Centre at the University of Cambridge. From 2002 to 2005, he held a Canada Research Chair in the Department of Materials Engineering at the University of British Columbia, Vancouver. He is widely known in the gas turbine community for his work on the physical metallurgy of the superalloys, and has taught extensively in this field.

The Superalloys Fundamentals and Applications

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But if I were to say, my fellow citizens, that we shall send a rocket to the moon, 240 000 miles away from the control station in Houston, a giant rocket more than three hundred feet tall, the length of this football field, made of new metal alloys, some of which have not yet been invented, capable of standing heat and stresses several times more than have ever been experienced, fitted together with a precision better than the finest watch, carrying all the equipment needed for propulsion, guidance, control, communications, food and survival, on an untried mission, to an unknown celestial body, and then return it safely to Earth, re-entering the atmosphere at speeds of over 25 000 miles per hour, causing heat about half that of the temperature of the sun, almost as hot as it is here today, and do all this, and do it right, and do it first before this decade is out, then we must be bold . . .

From John F. Kennedy's speech at Rice University, Houston, Texas, 12 September 1962

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Foreword

I am grateful to Cambridge University Press for allowing me to make some comments about Roger Reed's new textbook *The Superalloys: Fundamentals and Applications*. Nickel-based superalloys represent a very important class of engineering material, finding widespread application for example in critical components within the gas turbine engines used for jet propulsion and electricity generation. This is due to their superior mechanical properties that are maintained to elevated temperatures. Indeed, new classes of superalloy are continually being sought by gas turbine manufacturers around the world for applications in the hottest parts of the engine. This is because higher temperatures result in improvements to the efficiency of the engine and therefore lower fuel burn. Engine performance is a major factor in any power plant competition, which helps to explain why all the engine manufacturers spend so much money developing future generations of superalloys.

The author has provided us with a textbook covering both the fundamentals and applications of superalloy technology. This is a significant and unique achievement, especially given the broad range of subject matter dealt with. In Chapter 1, the requirement for materials capable of operating at elevated temperatures is introduced along with the historical development of the nickel-based superalloys and their emergence as materials for high-temperature applications. Chapter 2 concerns the physical metallurgy of the superalloys, with an emphasis on the details which distinguish them from other classes of engineering alloys, for example the gamma prime strengthening phase, the role of defects such as the anti-phase boundary, the unique particle-strengthening mechanisms, the anomalous yield effect and the creep deformation behaviour. Chapters 3 and 4 deal with superalloy technology as applied to two major gas turbine components of critical importance: the turbine blade and discs. Of note is the balanced coverage given to the processing, alloy design and microstructure/property relationships relevant to superalloys used for these distinct applications. Chapter 5 deals with surface coatings technologies, which are becoming increasingly critical as operating temperatures continue to rise. In Chapter 6, projections are made about the future of superalloy technology.

In view of the author's exemplary treatment of the subject, this textbook deserves to become the definitive textbook in the field for the foreseeable future. I have found myself referring to it on many occasions already! It is recommended to all those with an interest in the field of high-temperature materials, particularly those involved with gas turbine technology or those embarking on a higher degree in the subject. Also, since the superalloys

represent a considerable success story in the field of materials science and engineering, it is recommended for use within the materials-related curricula at universities.

Dr Mike Hicks
Chief Technologist – Materials
Rolls-Royce plc

Preface

Based upon nickel, but containing significant amounts of at least ten other elements including chromium and aluminium, the superalloys are high-temperature materials which display excellent resistance to mechanical and chemical degradation at temperatures close to their melting points. Since they first emerged in the 1950s, these alloys have had a unique impact. Consider the aeroengines which power the modern civil aircraft. The superalloys are employed in the very hottest sections of the turbines, under the heaviest of loads, with the utmost importance placed on assuring the integrity of the components fabricated from them. Indeed, the development of the superalloys has been intrinsically linked to the history of the jet engine for which they were designed; quite simply, a modern jet aeroplane could not fly without them. Further improvements in temperature capability are now being actively sought, for example for the engines to power the two-decked Airbus A380 and the Boeing 787 Dreamliner. Superalloys are being employed increasingly in the land-based turbine systems used for generating electricity, since fuel economy is improved and carbon emissions are reduced by the higher operating conditions so afforded. But new developments in superalloy metallurgy are required for the next generation of ultra-efficient power generation systems. Over the next 25 years, the world's installed power generation capacity is expected to double, due to the rapidly growing economies and populations of the developing countries, and because most of the current plant in the developed countries will need to be replaced. Thus the superalloys have never been more important to the world's prosperity.

The remarkable performance of today's superalloys is not merely fortuitous. Numerous researchers and technologists have worked to develop the basic understanding of their physical behaviour and the more practical aspects required to put these alloys to best use. With this book, the reader is presented with an introduction to the metallurgical principles which have guided their development. It turns out that the topics of alloy design, process development, component engineering, lifetime estimation and materials behaviour are very closely inter-related. The book is aimed at those pursuing degrees in Materials Science and Engineering, but those studying Mechanical Engineering, Aerospace Engineering, Physics and Chemistry will also find it useful. It has been developed with two audiences in mind: first, final-year undergraduate students taking an elective course in high-temperature materials technology, and, secondly, students embarking on a higher degree in this field who seek an introduction to the subject. Included at the end of the first five chapters are questions designed to test the reader. Many of these require numerical working using a calculator, spreadsheet or computer programming tool. Most of these exercises have been tested on

students at Imperial College, the University of Cambridge and the University of British Columbia, where I have held teaching positions. I would like to receive comments from those who have attempted the problems – I will attempt to answer all queries as promptly as I can. My email address can be found at www.cambridge.org/9780521859042.

In preparing the book, the reader should be aware that it has been necessary to be selective about the content included in it. I have included sufficient material to satisfy the requirements of a one-course semester – about 30 hour-long periods, some of which should be used for classwork, problem sets and design exercises. In my own teaching I have used the book's contents to emphasise the relationships between the structure/composition of these materials, their mechanical/chemical behaviour, the processing of them and the design of the components which have provided the technological impetus for their development. Thus, a proper balance between these different topics has been sought; the need for this makes the superalloys an excellent case study in the field of materials.

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1 Introduction

1.1 Background: materials for high-temperature applications

1.1.1 Characteristics of high-temperature materials

Certain classes of material possess a remarkable ability to maintain their properties at elevated temperatures. These are the *high-temperature materials*. Their uses are many and varied, but good examples include the components for turbines, rockets and heat exchangers. For these applications, the performance characteristics are limited by the operating conditions which can be tolerated by the materials used. For example, the thrust and fuel economy displayed by the modern aeroengine is strongly dependent upon, and limited by, the high-temperature strength of the nickel-based superalloys used for its hottest sections.

What are the desirable characteristics of a high-temperature material? The first is *an ability to withstand loading at an operating temperature close to its melting point*. If the operating temperature is denoted T_{oper} and the melting point T_{m} , a criterion based upon the homologous temperature τ defined as $T_{\text{oper}}/T_{\text{m}}$ is sensible; this should be greater than about 0.6. Thus, a superalloy operating at 1000 °C in the vicinity of the melting temperature of nickel, 1455 °C, working at a τ of $(1000 + 273)/(1455 + 273) \sim 0.75$, is classified as a high-temperature material. But so is ice moving in a glacier field at -10 °C, since τ is $263/273 \sim 0.96$, although its temperature is substantially lower. A second characteristic is *a substantial resistance to mechanical degradation over extended periods of time*. For high-temperature applications, a time-dependent, inelastic and irrecoverable deformation known as *creep* must be considered – due to the promotion of thermally activated processes at high τ . Thus, as time increases, creep strain, ϵ_{creep} , is accumulated; for most applications, materials with low rates of creep accumulation, $\dot{\epsilon}_{\text{creep}}$, are desirable. In common with other structural materials, the *static* properties of yield stress, ultimate tensile strength and fracture toughness are also important – and these must be maintained over time. A final characteristic is *tolerance of severe operating environments*. For example, the hot gases generated in a coal-fired electricity-generating turbine are highly corrosive due to the high sulphur levels in the charge. Kerosene used for aeroengine fuel tends to be cleaner, but corrosion due to impurities such as potassium salts and the ingestion of sea-water can occur during operation. In these cases, the high operating temperatures enhance the possibility of oxidation. Under such conditions, any surface degradation reduces component life.

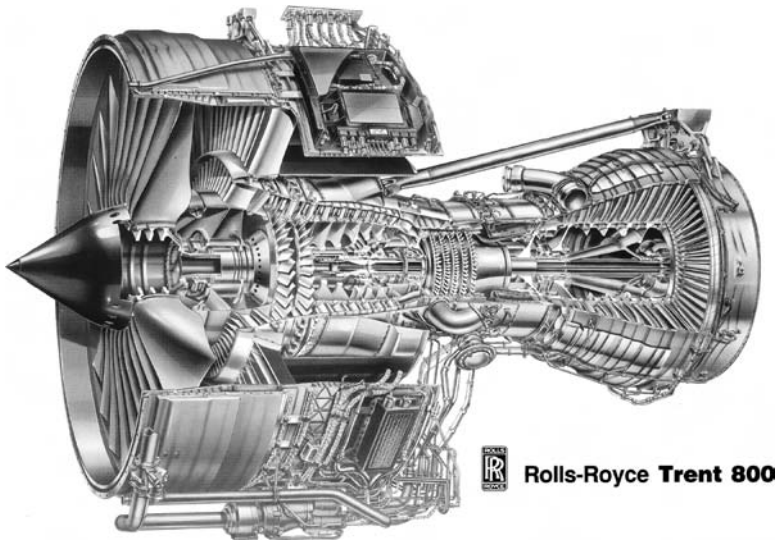


Fig. 1.1. Artist's impression of the turbomachinery in Rolls-Royce's Trent 800 engine, which powers the Boeing 777 aircraft. (Courtesy of Rolls-Royce.)

1.1.2 *The superalloys as high-temperature materials*

When significant resistance to loading under static, fatigue and creep conditions is required, the nickel-base superalloys [1–3] have emerged as the materials of choice for high-temperature applications. This is particularly true when operating temperatures are beyond about 800 °C. This is the case for gas turbines used for jet propulsion, for example, the 100 000 lb thrust engines used for the Rolls-Royce Trent 800 and General Electric GE90 which power the Boeing 777 [4] (see Figure 1.1), but also the smaller 1000 lb engines used for helicopter applications. Gas turbines are used also for electricity generation, for example, the 250 MW gas-fired industrial plant which can generate enough power to satisfy a large city of a million people (see Figure 1.2), or the smaller 3 MW gas-fired generators suitable for back-up facilities [5]. When weight is a consideration, titanium alloys are used, but their very poor oxidation resistance restricts their application to temperatures below about 700 °C [6]. For some electricity-generating power plant applications which rely upon superheated steam at 565 °C, high-strength creep-resistant ferritic steels are preferred on account of their lower cost. However, the latest generation of ultra-supercritical steam-generating coal-fired power stations requires boiler tubing that can last up to 200 000 hours at 750 °C and 100 MPa – new types of superalloy are being developed for these applications [7], since ferritic steels cannot be designed to meet these property requirements. Generally speaking, ceramics such as silicon carbide and nitride are not used for these applications, despite their excellent oxidation and creep resistance, due to their poor toughness and ductility. Zirconia-based ceramics do, however, find applications in thermal barrier coatings which are used in association with the superalloys for high-temperature applications.

Since the technological development of the superalloys is linked inextricably to the gas turbine engine, it is instructive to consider the functions of its various components [8].



Fig. 1.2. Artist's impression of the turbomachinery in Siemens Westinghouse's W501F gas turbine engine, used for electricity generation. (Courtesy of Siemens Westinghouse.)

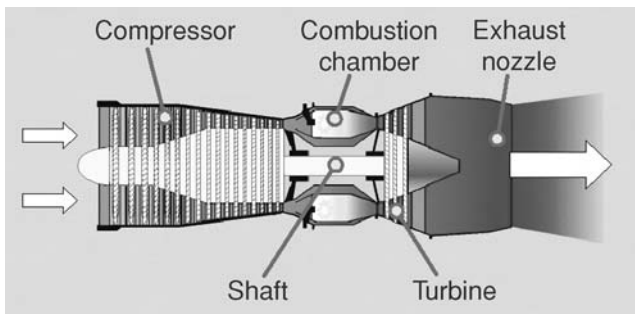


Fig. 1.3. Diagram illustrating the basic features of a very basic gas turbine engine: the turbojet. (Courtesy of Rolls-Royce.)

Consider the turbojet engine, Figure 1.3. The role of the *compressor*, consisting of compressor blades and discs, is to squeeze the incoming air, thus increasing its pressure. The compressed air enters the *combustor*, where it is mixed with fuel and ignited. The hot gases are allowed to expand through a *turbine*, which extracts the mechanical work required to drive the compressor; this necessitates a *shaft*, which transmits the torque required for this to happen. In the case of the turbojet, thrust arises from the momentum change associated with the incoming air being accelerated and its emergence as exhaust gas at a significantly higher velocity. Variants of this basic design are possible. In a turboprop engine, an additional low-pressure compressor, or *fan*, is added to the front of the engine. Although this necessitates an additional shaft and an associated turbine to drive it, the weight penalty associated with the extra turbomachinery is offset by the greater fuel economy – this arises from the

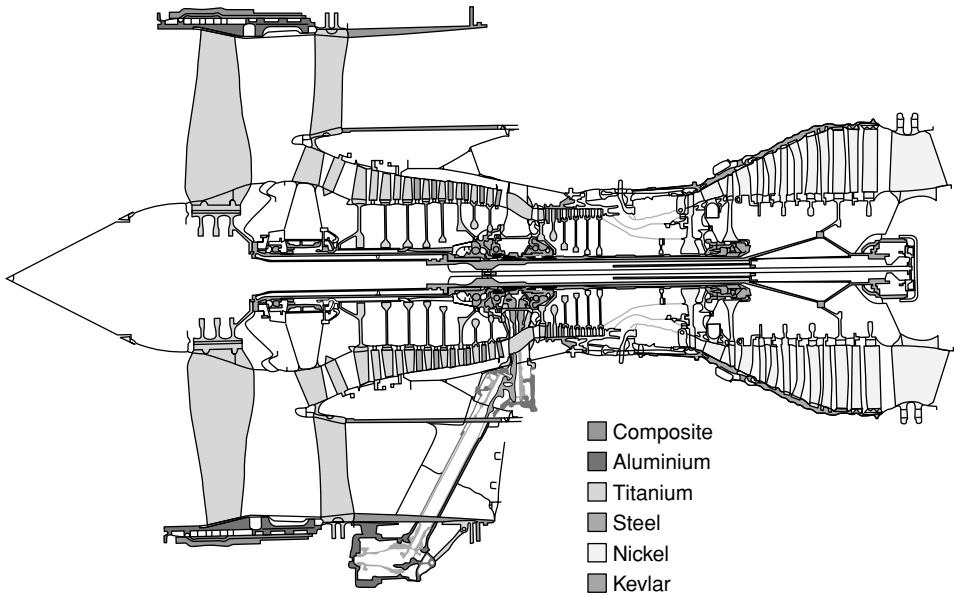


Fig. 1.4. Illustration of material usage in the Trent 800 engine. Note the extensive use of nickel-based superalloys in the combustor and turbine sections (see the back cover for a colour version of this figure). (Courtesy of Rolls-Royce.)

thrust provided from the air, which *bypasses* the turbines. Figure 1.4 illustrates the different materials used in the various parts of the Trent 800 aeroengine. One sees that titanium alloys are chosen for the fan and compressor sections, on account of their low density, good specific strength and fatigue resistance. However, in the combustor and turbine arrangements, the nickel-based superalloys are used almost exclusively. Superalloys are used also in the final (high-pressure) stages of the compressor.

When designing a gas turbine engine, great emphasis is placed on the choice of the *turbine entry temperature* (TET): the temperature of the hot gases entering the turbine arrangement [9]. There, the temperature falls as mechanical work is extracted from the gas stream; therefore the conditions at turbine entry can be considered to be the most demanding on the turbomachinery and the nickel-based superalloys from which are they made. As will be seen in Section 1.2, the performance of the engine is greatly improved if the TET can be raised – and over the 50 years since their conception, this has provided the incentive and technological impetus to enhance the temperature capability of the superalloys, and to improve their processing and the design of components fabricated from them. The success of these enterprises can be judged from the way in which the TET of the large civil aeroengine has increased since Whittle's first engine of 1940 (Figure 1.5); a 700 °C improvement in a 60-year period has been achieved [10]. Of course, the TET varies greatly during a typical flight cycle (see Figure 1.6), being largest during the take-off and climb to cruising altitude. Turbines for power-generating applications experience fewer start-up/power-down cycles but very much longer periods of operation, during which the TET tends to be rather constant.

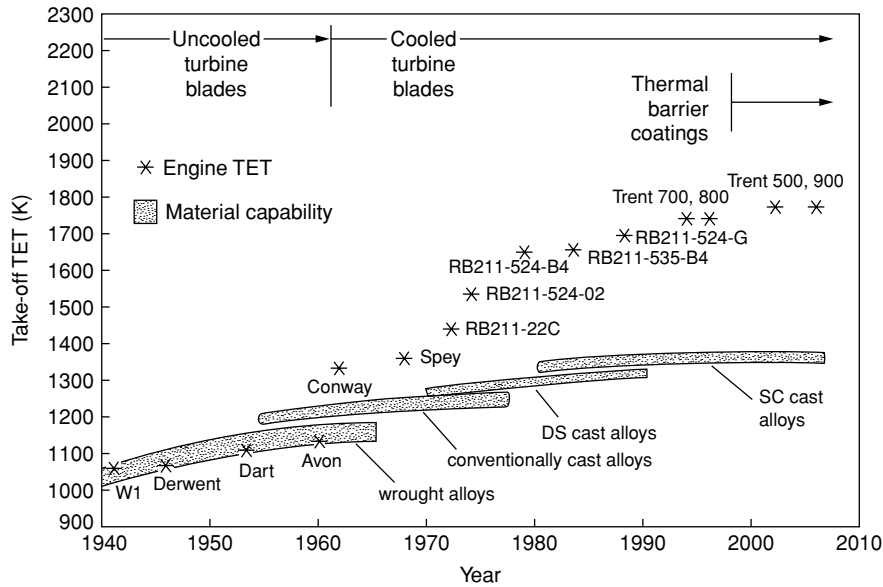


Fig. 1.5. Evolution of the turbine entry temperature (TET) capability of Rolls-Royce's civil aeroengines, from 1940 to the present day. Adapted from ref. [10].

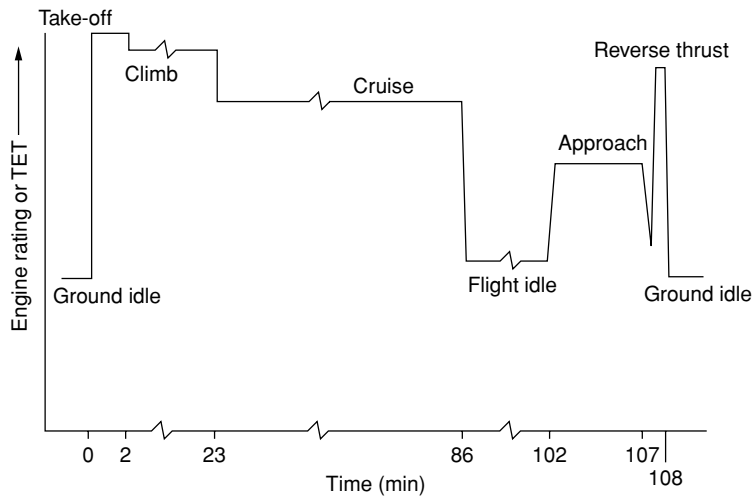


Fig. 1.6. Variation of the turbine entry temperature (TET) during a typical flight cycle of a civil aircraft.

1.1.3 Instances of superalloy component failures

The failure of components fabricated from the superalloys is a rare occurrence, since the applications demand very conservative designs and significant safety factors. Nevertheless, failures have occurred and undoubtedly there will be further ones in the future. Examination of some examples emphasises a number of important points.

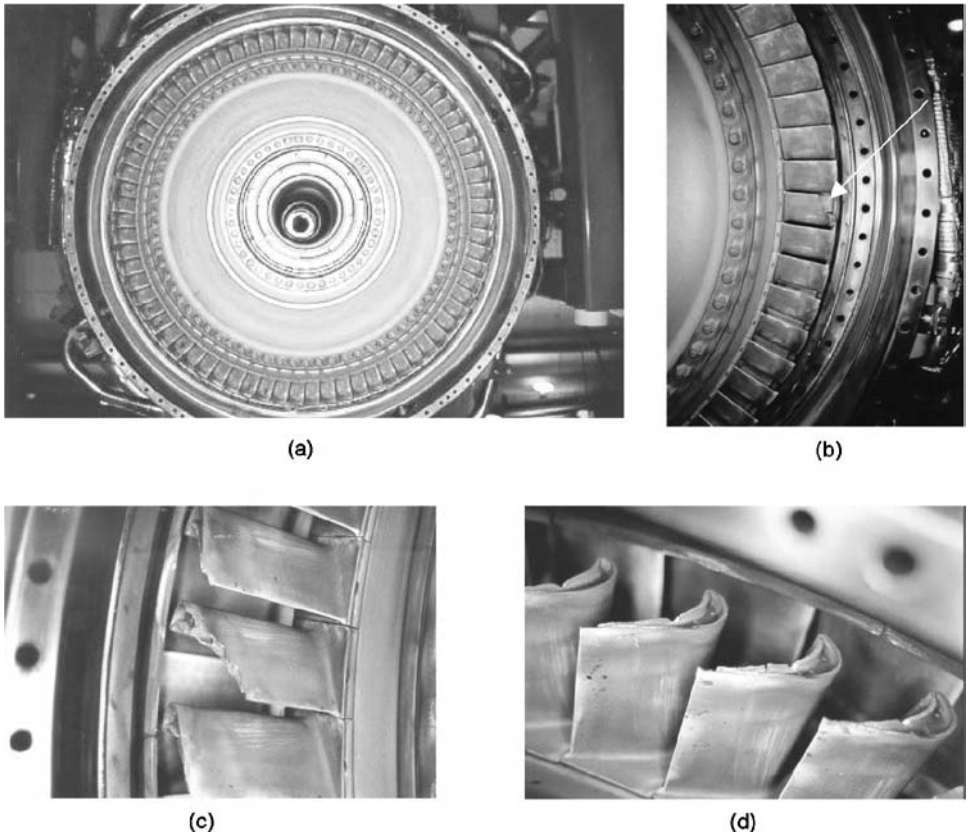


Fig. 1.7. Images of the General Electric CFM56 turbofan engine, which failed in service in October 2000, shortly after taking off from Hobart, Australia [11]. (a) General view of the high-pressure turbine; (b) high-pressure turbine rotor with the failed blade indicated; (c) failed high-pressure turbine blade; (d) tip notch damage observed on the remaining blades.

Consider first the case of a Boeing 737 powered by two General Electric CFM56 turbofan engines, which, on 13 October 2000, experienced an in-flight engine failure that resulted in the engine being shut down and the aircraft returning safely to Hobart airport in Tasmania, Australia [11]. After disassembly and inspection of the engine, failure was attributed to the loss of a 15 mm by 20 mm segment from the trailing edge of a single high-pressure (HP) turbine blade (see Figure 1.7) fabricated from the Rene 125 alloy. This passed into the low-pressure turbine stages, where it caused overloading and collapse of the entire blade array. Metallographic examination of the HP blades indicated that radial cracks near the blade tips were common, as a result of severe thermal cycling, high thermal gradients and thermal fatigue. Many of the tips had been weld-repaired using the Rene 80 alloy. In the failed blade, tip cracks had grown into a v-shaped notch because of oxidation and corrosion effects, and had intercepted a deep underlying repair weld fabricated from the lower strength superalloy Inconel 625. On reaching the base of the repair weld, the resultant fatigue stresses were sufficient to propagate the cracking to the point of final failure. The failed

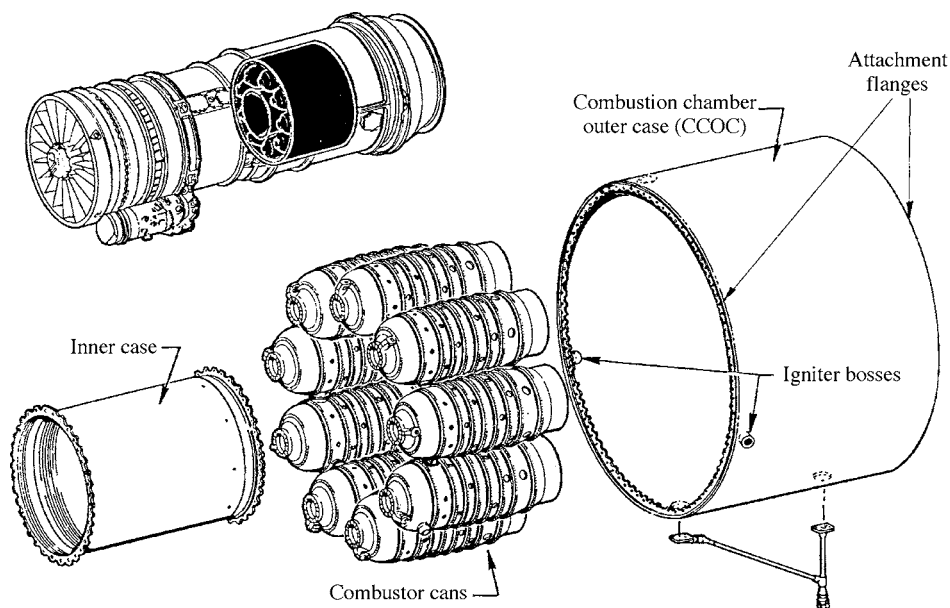


Fig. 1.8. Schematic diagram of the arrangement of the combustor in the Pratt & Whitney JT8D-15 engine [12].

blade had completed 17 928 flight cycles, and had flown 5332 cycles since its repair and overhaul.

A second example had very much more tragic consequences. On 22 August 1985, at Manchester Airport in the United Kingdom, a British Airways Boeing 737 carrying 131 passengers and 6 crew suffered an uncontained engine failure during take-off, which was therefore aborted by the crew [12]. Unfortunately, pieces from the port Pratt & Whitney JT8D-15 engine were ejected from the engine – and these punctured a fuel tank causing a catastrophic fire, in which 55 persons on board lost their lives. The aeroplane had not left the ground. The origin of the failure was a 360° separation of the No 9 combustor can (see Figure 1.8), which consisted of 11 pieces of the superalloy Hastelloy X in sheet form, welded together; this allowed hot gases to escape from it and impinge upon the inner surface of the combustion chamber outer case, which ruptured catastrophically during take-off due to localised overheating, see Figure 1.9. In November 1983, after 3371 cycles, the No 9 can had been inspected and circumferential cracking of 180 mm combined length had been repaired, by fusion welding. Solutioning and weld stress-relief heat treatments had not been applied. It lasted a further 2036 cycles before failure.

In these instances, lessons should be learned from these tragic circumstances. The following should be clear to the reader. First, the modern jet engine relies very heavily upon the superalloys to withstand the significant loads and temperatures developed during operation. Secondly, where defects exist due to manufacturing or else due to damage accumulated during service, these can lead to catastrophic failure; hence, for these applications it is absolutely critical to ensure the components fabricated from the superalloys are of the highest

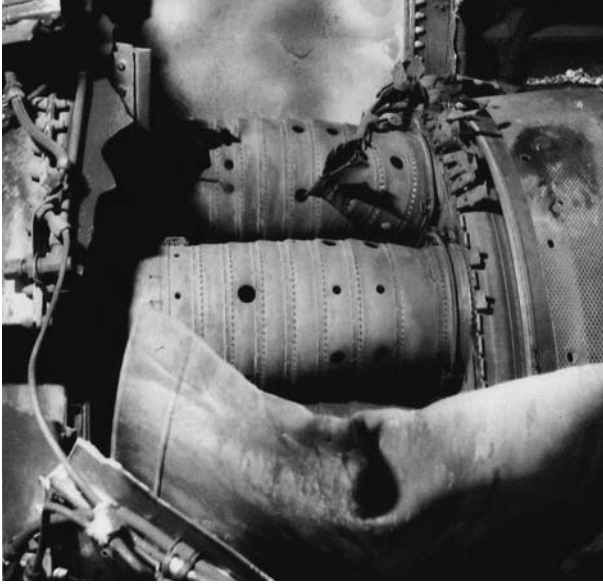


Fig. 1.9. View of failed combustor in the failed Pratt & Whitney JT8D-15 engine, August 1985, Manchester Airport, UK [12].

integrity and quality. Thirdly, procedures for inspection and repair are critical if ongoing operation is to remain safe. Finally, a close integration of procedures for component design, manufacturing and lifetime estimation is required in order to ensure the safe working conditions of components fabricated from the superalloys, and gas turbines in general.

1.2 The requirement: the gas turbine engine

The gas turbine used for jet propulsion and electricity generation is an example of a heat engine. The power output therefore depends upon the temperature increment through which the working fluid is raised – it is improved if this can be increased.

In order to reinforce this concept, consider Carnot's reversible heat engine, which consists of an ideal gas that cycles between hot and cold reservoirs of temperatures T_1 and T_2 , respectively; see Figure 1.10(a). Once each cycle, an amount of heat, Q_{AB} , is absorbed from the hot reservoir and a quantity, Q_{CD} , is transferred to the cold one; a fraction, η , of the heat, Q_{AB} , absorbed from the hot reservoir is converted into useful work, W , such that the efficiency, η , equals W/Q_{AB} with conservation of energy requiring that $W = Q_{AB} - Q_{CD}$.

It is relatively easy to determine expressions for η and W . Writing the ideal gas equation as $pV = nRT$, where p is pressure, V is volume, T is temperature, R is the gas constant and n is the number of moles, one must consider the following four steps [13].

- (1) The gas, in perfect thermal contact with the hot reservoir, undergoes an infinitely slow expansion from volume V_A to V_B . During this process, the work done by the gas, W_{AB} ,

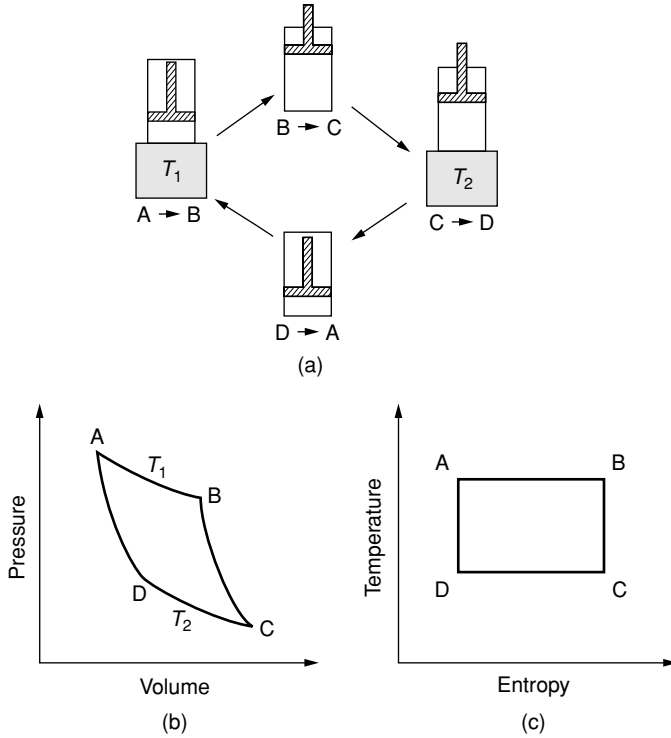


Fig. 1.10. The Carnot cycle for an ideal gas. (a) The four steps of the cycle: two isothermal and two adiabatic. (b) Corresponding pressure–volume diagram. (c) The associated temperature–entropy diagram. Adapted from ref. [13].

is given by

$$W_{AB} = \int_{V_A}^{V_B} p \, dV = \int_{V_A}^{V_B} \frac{nRT_1}{V} \, dV = nRT_1 \ln \left\{ \frac{V_B}{V_A} \right\} = Q_{AB} \quad (1.1)$$

- (2) A second step, in which thermal contact to the reservoir is broken and the gas is insulated from its surroundings: the gas undergoes an adiabatic expansion from state B to state C along the adiabat BC, such that $p_B V_B^\gamma = p_C V_C^\gamma$, where γ is the ratio of specific heats at constant pressure and volume. Since the gas is isolated, $Q_{BC} = 0$. The work done by the gas, W_{BC} , is given by

$$W_{BC} = \int_{V_B}^{V_C} p \, dV = \int_{V_B}^{V_C} \frac{p_B V_B^\gamma}{V^\gamma} \, dV = \frac{p_C V_C - p_B V_B}{1 - \gamma} = \frac{nR(T_1 - T_2)}{\gamma - 1} \quad (1.2)$$

where T_1 and T_2 are the initial and final temperatures, respectively.

- (3) A third step: the gas, in contact with the cold reservoir at temperature T_2 , undergoes an isothermal compression from V_C to V_D . Work is done on the gas such that

$$W_{CD} = \int_{V_C}^{V_D} p \, dV = \int_{V_C}^{V_D} \frac{nRT_2}{V} \, dV = nRT_2 \ln \left\{ \frac{V_D}{V_C} \right\} = Q_{CD} \quad (1.3)$$

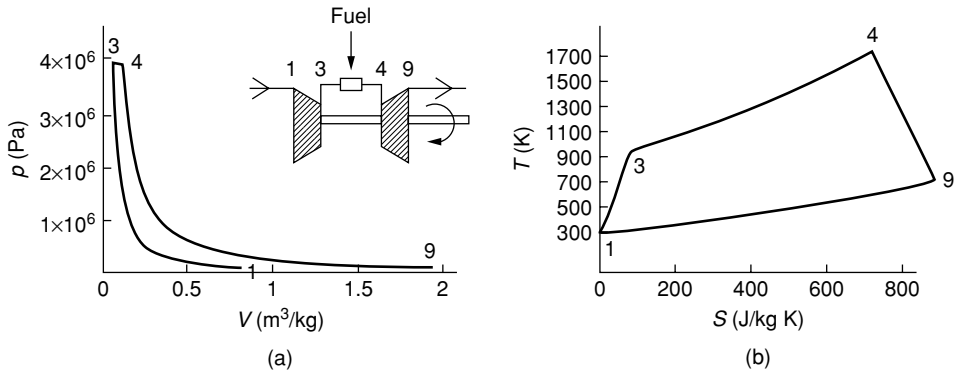


Fig. 1.11. Pressure–volume and temperature–entropy diagrams for the turbojet. Adapted from ref. [10].

- (4) In the final step, an adiabatic compression takes the gas from state D to its initial state A. Appealing to the result of step 2, the work done *on* the gas is given by

$$W_{DA} = \frac{nR(T_2 - T_1)}{\gamma - 1} \quad (1.4)$$

Since the gas is isolated, $Q_{DA} = 0$. Note that $W_{DA} = -W_{BC}$.

The total work, W , obtained in the cycle is given by

$$W = W_{AB} + W_{BC} + W_{CD} + W_{DA} = Q_{AB} + Q_{CD} = nRT_1 \ln \left\{ \frac{V_B}{V_A} \right\} - nRT_2 \ln \left\{ \frac{V_C}{V_D} \right\} \quad (1.5)$$

and is equal to the area enclosed by the loop ABCD in Figure 1.10(b); this follows from a consideration of the four integrals, $\int p dV$, defining W . The efficiency, η , defined as the ratio of the useful work obtained, W , to the heat absorbed from the heat source, Q_{AB} , is given by

$$\eta = \frac{W}{Q_{AB}} = 1 - \frac{T_2 \ln\{V_C/V_D\}}{T_1 \ln\{V_B/V_A\}} = 1 - \frac{T_2}{T_1} \quad (1.6)$$

where use has been made of relationships such as $p_A V_A = p_B V_B$ and $p_B V_B^\gamma = p_C V_C^\gamma$, etc. It follows that if T_1 is raised, or T_2 is reduced, the work extracted from the cycle is increased and the efficiency improved. In practice, raising T_1 is the more practical option – with the limit for T_1 being the capability of the turbomachinery to withstand the high-temperatures and stresses involved. Figure 1.10(c) shows the temperature–entropy plot for the reversible Carnot cycle.

The assumptions made in the Carnot cycle make it only a first approximation for the gas turbine engine, particularly as additions of mass and heat are made to the working fluid via the burning of fuel. However, the fuel-to-air ratio is typically only 0.02–0.03 at maximum power and 0.01 under cruise conditions. Therefore, the pressure–volume cycle for a simple turbojet (see Figure 1.11(a)) exhibits many of the features expected from the Carnot cycle; note in particular the 40-fold increase in pressure. Figure 1.11(b) shows the corresponding temperature–entropy plot (the numbers 1, 3, 4 and 9 have been chosen for

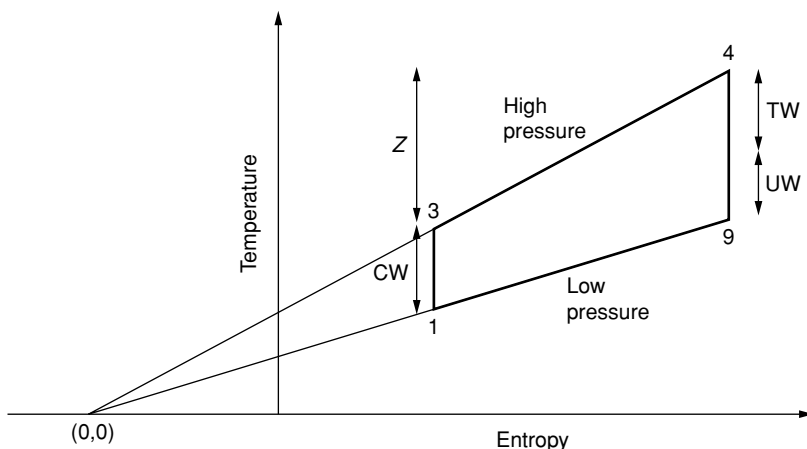


Fig. 1.12. Simplified temperature–entropy diagram for the turbojet, with compression and turbine stages assumed to be isentropic. CW = compression work; TW = turbine work; UW = useful work; Z = ideal work of combustion.

consistency with commonly used terminology in gas-turbine design). Note that neither the work of compression nor the work of expansion in the compressor are isentropic, implying irreversibility, and that the chemical energy introduced in the combustion chamber leads to a significant increase in temperature of about 800 K. These effects are not accounted for in the Carnot analysis.

To make approximate but quantitative calculations, one can assume that the compressor and turbine stages are isentropic, so that the lines 1–3 and 4–9 in Figure 1.11(b) are vertical. Furthermore, the lines 3–4 and 1–9 can be assumed to be straight but diverging; they must of course pass through the point (0, 0) since at zero kelvin gases have no entropy. The situation is then as depicted in Figure 1.12. The distance 1–3 is proportional to the compression work (CW), and the distance 4–9 is proportional to the sum of the turbine work (TW) and the useful work (EW): the work of expansion (EW). The vertical distance between 3 and 4 measured along the temperature axis is proportional to Z, the rate of addition of chemical energy from the fuel, or the ideal work of combustion. Because the lines 3–4 and 1–9 diverge, and since for an ideal cycle $CW = TW$, the quantity UW is finite and positive; it can be used for propulsion or else to drive an external shaft for electricity generation.

Example question

A turbojet is operating with a TET of 1400 K, an overall pressure ratio (OPR) of 25 and an air-mass intake, $\dot{M}$, of 20 kg/s. Assuming an ideal cycle such that the compression work equals the turbine work, determine the rate of useful working (or specific power) available to provide a propulsive force. Hence estimate the thermal efficiency, η_{th} . Assume adiabatic, reversible processes where necessary. As a further exercise, estimate (i) the thrust, F_N , developed and (ii) the propulsive efficiency, η_{prop} , if the velocity of the intake air is 250 m/s. (Assume $c_p = 1000 \text{ J/(kg K)}$, $\gamma = 1.4$ and the intake air to have a temperature of 288 K and velocity V_0 . Take atmospheric pressure to be 101 325 Pa.)

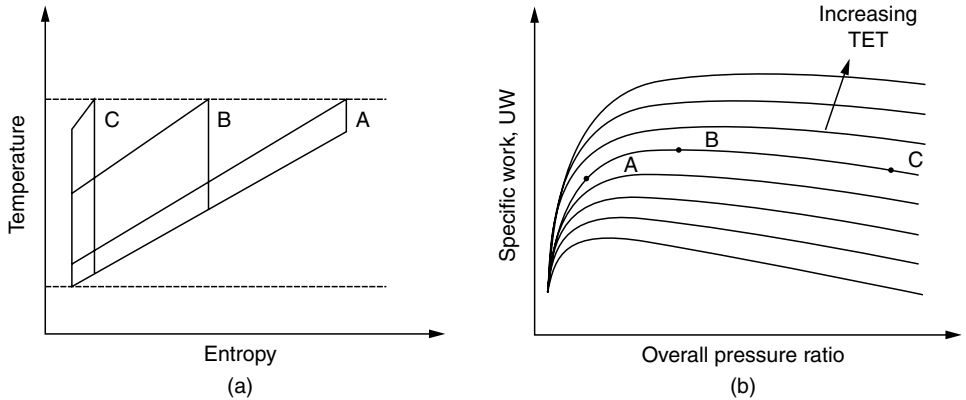


Fig. 1.13. (a) Idealised temperature–entropy diagrams for the turbojet, for large (C), medium (B) and small (A) overall pressure ratios. (b) Plots of specific work vs. OPR for various TETs.

Solution For an adiabatic, reversible process one has

$$\frac{T_3}{T_1} = \left(\frac{p_3}{p_1} \right)^{\frac{\gamma-1}{\gamma}}$$

and since the pressure ratio $p_3/p_1 = 25$, this yields $T_3 = 723$ K if $T_1 = 288$ K. In a similar way, $T_9 = 558$ K if $T_4 = 1400$ K. Then $Z = \dot{M}c_p(T_4 - T_3) = 13.6$ MW, $CW = TW = \dot{M}c_p(T_3 - T_1) = 8.7$ MW and $EW = \dot{M}c_p(T_4 - T_9) = 16.8$ MW. It follows that the useful work, UW, is $16.8 - 8.7 = 8.1$ MW.

The thermal efficiency is defined as the ratio of useful work to the work of combustion, i.e.

$$\eta_{th} = UW/Z = 8.1/13.6 = 0.60$$

These considerations demonstrate that the turbine entry temperature, T_4 , needs to be as high as possible, if the useful work is to be maximised. Since the constant pressure lines diverge, UW is greatest when T_4 is large. There does, however, need to be a balance between a high TET and a large enough OPR; see Figure 1.13. This is because at very low or very high OPRs, the *difference* between CW and the EW is minimal, and, since $CW = TW$, then negligible work is available for propulsion.

The useful work, UW, is spent increasing the kinetic energy of the gas stream passing through the engine; hence,

$$UW = \frac{1}{2} \dot{M} [V_{jet}^2 - V_0^2]$$

so that $V_{jet} = 934$ m/s. The thrust, F_N , is then given by (from momentum considerations)

$$F_N = \dot{M} [V_{jet} - V_0]$$

or $20 \times (934 - 250) = 13\,680$ N.

The propulsive efficiency is a measure of the proportion of jet kinetic energy (relative to the engine) which is converted to rate of working (by the thrust) on the aeroplane, defined

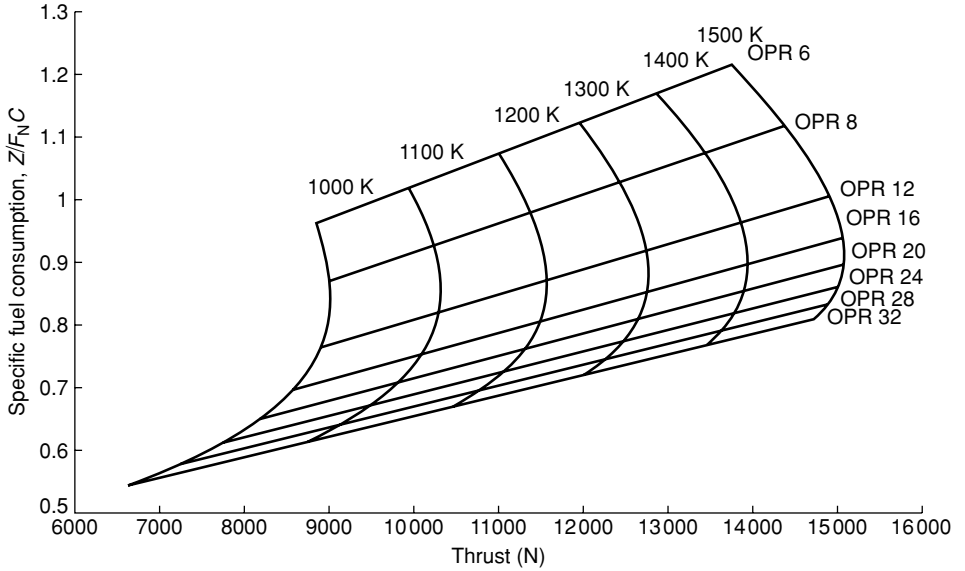


Fig. 1.14. Performance diagram illustrating the variation of the specific fuel consumption (defined as normalised fuel burn/thrust developed) against thrust, for various values of the TET and OPR.

according to

$$\eta_{\text{prop}} = \frac{F_N V_0}{\frac{\dot{M}}{2}(V_{\text{jet}}^2 - V_0^2)}$$

This yields a value of 0.42.

Such calculations, although simplistic, can be used to justify the use of the bypass design for a modern civil aeroengine; see Figure 1.1. Thrust is required for propulsion, and this depends upon both η_{th} and η_{prop} . A high η_{th} demands large TET and reasonable OPR. Hence, if the available useful work, UW, is used primarily to accelerate the air passing through the engine, then it is inevitable that the jet velocity will be very high, and the thrust, F_N , and specific fuel consumption (SFC), too large. This is demonstrated in Figure 1.14. The propulsive efficiency, η_{prop} , is correspondingly very low. The only way around this dilemma is to use a bypass design. Most of the useful work developed by the core of the engine is used to accelerate a much greater mass of air through a fan to a modest exit velocity. The remaining useful work is used to accelerate the core air to a rather modest V_{jet} .

These considerations explain why a modern civil turbofan aeroengine has a bypass ratio between 6:1 and perhaps 9:1 – to develop reasonable η_{prop} and SFC. For a military aeroengine, however, a low bypass ratio of 1.2:1 is more appropriate since a high thrust-to-weight ratio is required; furthermore, the fan diameter must be small to prevent the engine protruding from the airframe so that speed and acceleration is maximised. For further reading, refer to ref. [10] for a thorough and lucid introduction to the aerodynamic design of modern engines for civil and military aircraft. The analysis also explains why there has been a tremendous drive to increase both the pressure ratio and the turbine entry

temperature of large land-based industrial gas turbines, with values at the time of writing of 20 and 1425 °C, respectively, being commonplace; see Table 1.1 [14]. In turn, these more demanding operating conditions place more stringent requirements on the materials used. Table 1.2 summarises the alloys and coatings used in the hot sections of the latest generation of large IGTs [14]. As will become apparent, for IGT applications there is a trend towards (i) the use of single-crystal superalloys for the blading, with cobalt-based superalloys such as ECY-768 finding less favour and (ii) the application of coatings technologies for the provision of environmental protection. These topics are dealt with in detail in Chapters 3 and 5, respectively.

1.3 The selection of materials for high-temperature applications

Materials for high-temperature service must withstand considerable loads for extended periods of time. What are the best materials to choose for these applications? Can we justify the use of the superalloys which have nickel as the major constituent?

1.3.1 Larson–Miller approach for the ranking of creep performance

Resistance to creep deformation is a major consideration. For many materials and under loading conditions which are invariant with time, the creep strain rate, $\dot{\epsilon}_{ss}$, is constant; i.e. it approaches a steady-state [14]. This implies a balance of creep hardening, for example, due to dislocation multiplication and interaction with obstacles, and creep softening, for example, due to dislocation annihilation and recovery. Very often, it is found that

$$\dot{\epsilon}_{ss} = A\sigma^n \exp \left\{ -\frac{Q}{RT} \right\} \quad (1.7)$$

where σ is the applied stress, n is the stress exponent, A is a constant and Q is an activation energy. When a value for Q is deduced from the experimental creep data, one often finds that it correlates with the *activation energy for self-diffusion*, see Figure 1.15 [15]. This implies that some form of mass transport on the scale of the microstructure is rate-controlling.

Design against creep usually necessitates a consideration of the *time to rupture*, t_r , which usually satisfies the so-called Monkman–Grant relationship

$$t_r \times \dot{\epsilon}_{ss} = B \quad (1.8)$$

where B is a constant which is numerically equal to the creep ductility, i.e. the creep strain to failure. Then at constant σ one has

$$t_r \exp \left\{ -\frac{Q}{RT} \right\} = C \quad (1.9)$$

or

$$\log_{10} t_r - 0.4343 \frac{Q}{RT} = D \quad (1.10)$$

Table 1.1. *Evolution of the features of large land-based industrial gas turbines [5]*

Year of introduction	1967	1972	1979	1990 ^a	1998 ^b
Turbine inlet temperature (°C)	900	1010	1120	1260	1425
Pressure ratio	10.5	11	14	14.5	19–23
Exhaust temperature (°C)	427	482	530	582	593
Cooled turbine rows	R1 vane	R1, R2 vane, R1 blade	R1, R2 vane, R1, R2 blade	R1, R2, R3 vane, R1, R2, R3 blade	R1, R2, R3 vane, R1, R2, R3 blade
Power rating, MW	50–60	60–80	70–105	165–240	165–280
Efficiency, simple cycle (%)	29	31	34	36	39
Efficiency, combined cycle (%)	43	46	49	53	58

Note: ^a Corresponds approximately to GE 7F, 7A/9F, Westinghouse 501F/701F and Siemens V84.3/94.3 turbines.

^b Corresponds approximately to GE 7H/9H, Westinghouse 501G/701G, Siemens V84.3A/94.3A and ABB GT24/26 turbines.

Table 1.2. *Alloys and coatings used in the hot sections of various industrial gas turbines, as at the end of 1997 [5]*

Manufacturer	Model	Vanes	Blades	Coatings
ABB	11N2	IN939	IN738LC	NiCrAlY+Si
	GT24/26	DS CM247LC (R1)/ MarM247LC (R2,3)/ IN738 (R4,5)	DS CM247LC (R1-3)/ MarM247LC (R4,5)	TBC (R1V)/NiCrAlY+Si (R2-4 B,V) Uncoated R5V, Chromised (R5B)
GE	7/9EA	FSX-414 (all stages)	GTD-111/IN738/ Udimet 500	RT22/GT29-In+ (R1B)
	7/9FA	FSX-414 (R1)/ GTD-222 (R2,3)	DS GTD-111 (R1)/ GTD-111 (R2,3)	GT33-In/GT-29-In+/ Chromise (R3)
	7H	SC Rene N5 (R1)/ FSX-414/GTD-222	SC Rene N5 (R1)/ DS GTD-111 (R2,3)	TBC (R1,2 B,V)/ All others GT33
Siemens	V84/94.2	IN939	IN738LC/IN792 (R4)	CoNiCrAlY+Si
	V84/94.3A	SC PWA 1483 (R1,2)/ IN939	SC PWA 1483 (R1,2)	TBC (EB-PVD R1B)/MCrAlY+Re
Westinghouse/ Mitsubishi	501D5/701D	ECY-768/X-45	Udimet 520	MCrAlY
	501/701F	ECY-768/X-45	IN738LC	TBC (R1 B,V)/ MCrAlY/Sermalloy J
	501/701G	IN939	DS MarM002 (R1,2)/ CM247	TBC (R1,2 B,V)/ EB-PVD/MCrAlY

Note: R1, R2, etc. refer to the first, second rotor sections, etc.; B and V refer to blade and vane, respectively.

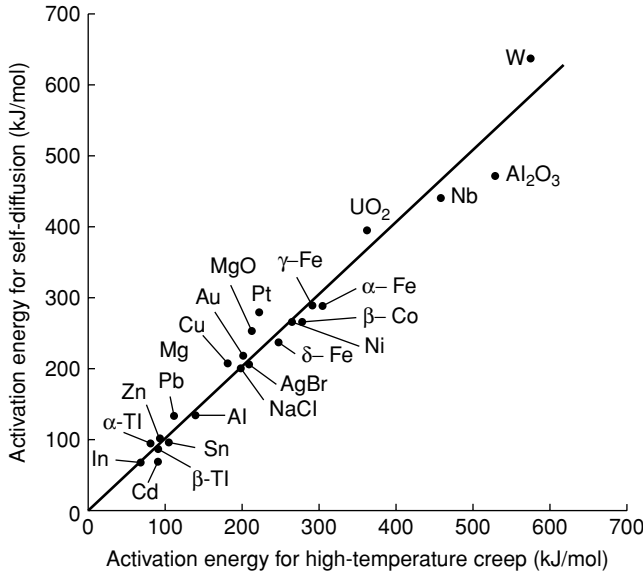


Fig. 1.15. Correlation between the creep activation energy and the energy for diffusion for several materials. Data taken from ref. [15].

where C and D are constants. Equation (1.10) can be written in the form

$$T[E + \log_{10} t_r] = P \quad (1.11)$$

where P is known as the Larson–Miller parameter and E is the Larson–Miller constant; this is found to vary between 15 and $25 \log h$ and is taken to be $20 \log h$. When creep tests for various combinations of (σ, T) are carried out, one usually finds a strong correlation between P and $\log \sigma$, consistent with the assumptions made. These plots are known as Larson–Miller diagrams.

In Figure 1.16, the performance of various materials are compared. Note that the alloys SRR99, CMSX-4 and RR3000 are first-, second- and third-generation single-crystal superalloys, which are used for the turbine blades of modern aeroengines. Waspaloy is a polycrystalline superalloy used for the turbine discs required to house the blading. MA754 is an oxide-dispersion-strengthened superalloy made by powder metallurgy. Ti-6242 is a titanium alloy used widely for the compressor sections of modern aeroengines; it has the highest creep resistance of any of the titanium alloys. Also shown are various intermetallic compounds such as NiAl, Ni₃Al, TiAl, Ti₃Al and FeAl, and other exotic alloys such as Nb-I, Mo-41Re and thoria-doped tungsten; these show promise as high-temperature materials, but they have not yet found application in the gas turbine. A number of points emerge from a consideration of Figure 1.16. First, for any given material the Larson–Miller line is approximately straight, although there is a tendency for some non-linearity, which results in the lines curving gently downwards. Second, in comparison with the other materials, it is clear that the single crystal superalloys behave excellently; the shift towards the right as one moves from SRR99, CMSX-4 to RR3000 has arisen due to the metallurgists who have been

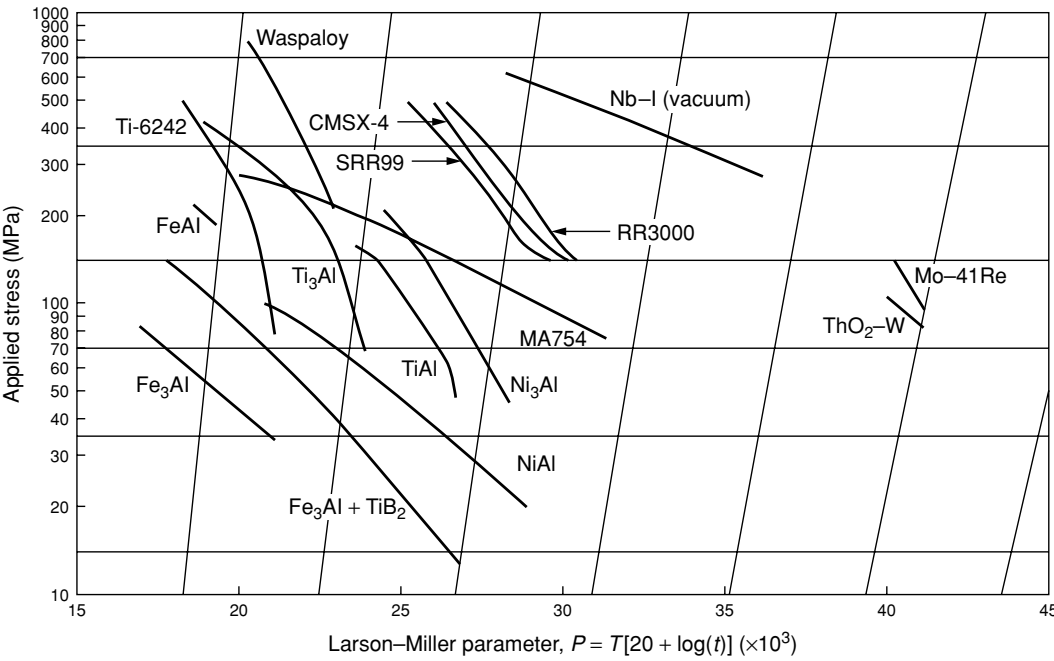


Fig. 1.16. Values of the Larson–Miller parameter, P , for a number of high-temperature materials. The horizontal and near-vertical lines have spacings equivalent to a factor of 2 change in creep life and 200°C temperature capability, respectively; the materials which display the best high-temperature performance lie towards the top right of the diagram. Adapted from a graph provided by Dan Miracle.

able to produce alloys of increasing creep resistance; these improvements turn out to be very substantial, although clearly if a step change could be made to a new alloy system, for example, one based upon niobium, then this would be very advantageous. Polycrystalline superalloys such as Waspaloy are less good in creep, but their behaviour exceeds that of many other alloy systems. The intermetallic compounds are not bad, particularly if one makes the comparison on a density-corrected basis; it turns out, however, that their toughness is no match for that displayed by the superalloys. Finally, titanium alloys behave well, although it has been found that their oxidation resistance beyond about $\sim 700^\circ\text{C}$ is very poor, and this limits their range of application – it has been found necessary, for example, to use superalloys for the last stages of the compressor sections of many aeroengines, even though titanium alloys are used for the first stages. Alternative materials based upon W or Mo show promise, although their density tends to be significant. Niobium alloys seem attractive, but their resistance to oxidation is very poor.

1.3.2 Historical development of the superalloys

Over the latter part of the twentieth century, a concerted period of alloy and process development enabled the performance of the superalloys to be improved dramatically. Although further improvements are being actively pursued, it is important to recognise this historical

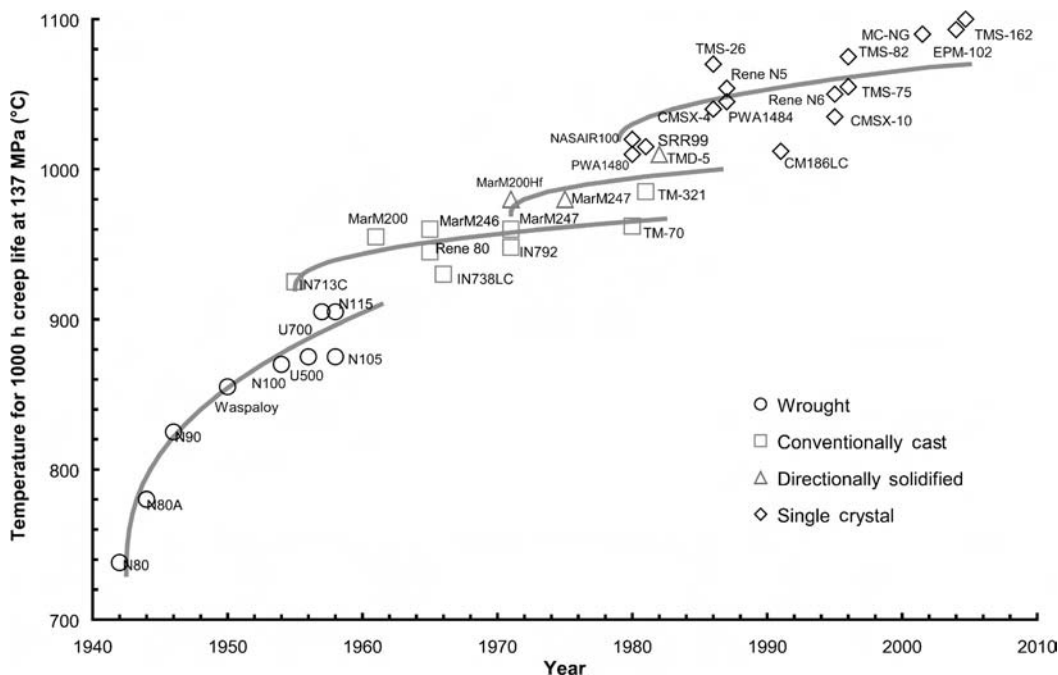


Fig. 1.17. Evolution of the high-temperature capability of the superalloys over a 60 year period since their emergence in the 1940s.

context, since much can be learned from it. Figure 1.17 provides a perspective for the alloy and process development which has occurred since the first superalloys began to appear in the 1940s; the data relate to the materials and processes for turbine blading, so that the creep performance (here taken as the highest temperature at which rupture occurs in not less than 1000 h, at 137 MPa) is a suitable measure for the progress which has been made. Various points emerge from a study of the figure. First, one can see that, *for the blading application*, cast rather than wrought materials are now preferred since the very best creep performance is then conferred. However, the first aerofoils were produced in wrought form. During this time, alloy development work – which saw the development of the first Nimonic alloys [1] – enabled the performance of the blading to be improved considerably; the vacuum induction casting technologies which were introduced in the 1950s helped with this since the quality and cleanliness of the alloys were improved dramatically. Second, the introduction of improved casting methods, and later the introduction of processing by directional solidification, enabled significant improvements to be made; this was due to the columnar microstructures that were produced in which the transverse grain boundaries were absent (see Figure 1.18). Once this development occurred, it was quite natural to remove the grain boundaries completely such that monocrystalline (single-crystal) superalloys were produced. This allowed, in turn, the removal of grain-boundary strengthening elements such as boron and carbon which had traditionally been added; this enabled better heat treatments to reduce microsegregation and eutectic content induced by casting, whilst avoiding incipient melting during heat treatment. The fatigue life is then improved.

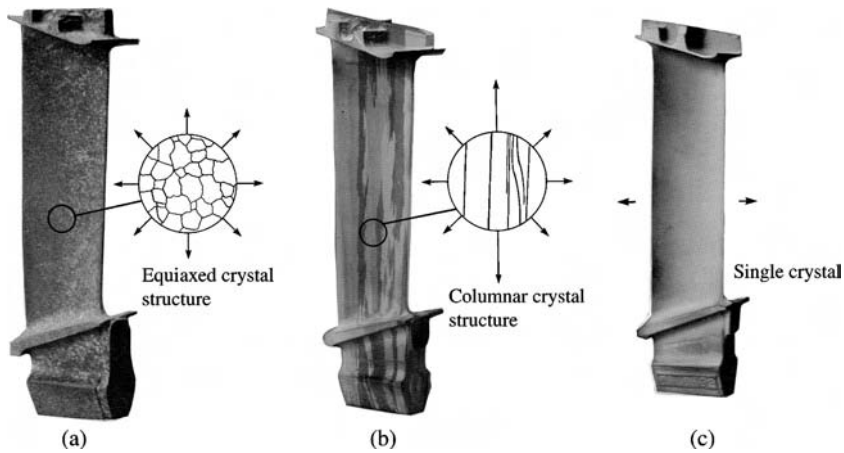


Fig. 1.18. Turbine blading in the (a) equiaxed, (b) columnar and (c) single-crystal forms [8].

Nowadays, single-crystal superalloys are being used in increasing quantities in the gas turbine engine; if the very best creep properties are required, then the turbine engineers turn to them, although it should be recognised that the use of castings in the columnar and equiaxed forms is still practised in many instances. It can be shown by the reader that the data in Figure 1.16 indicate that creep rupture lives of the single-crystal superalloys have been lengthened from about 250 h at 850 °C/500 MPa for a typical first-generation alloy such as SRR99, to about 2500 h for the third-generation alloy RR3000. Under more demanding conditions, for example, 1050 °C/150 MPa, rupture life has improved four-fold from 250 h to 1000 h. These improvements were won in a period of approximately 15 years between 1980 and 1995, primarily as a consequence of a better appreciation of the physical factors which confer high-temperature strength in these materials. It should be appreciated that each of the different engine manufacturers and materials suppliers have searched for their own proprietary compositions; Table 1.3 lists the compositions of common cast superalloys, including those used in single-crystal form. One can see that as many as 14 different alloying additions are added. The so-called first-generation single-crystal superalloys, such as PWA1480, Rene N4 and SRR99, contain appreciable quantities of the γ' hardening elements Al, Ti and Ta; the grain-boundary-strengthening elements C and B, which were added routinely to the earlier directionally solidified alloys, are no longer present. Second-generation alloys, such as PWA1484, CMSX-4 and Rene N5, are characterised by a 3 wt% concentration of Re, which is increased to about 6 wt% for the third-generation alloys such as CMSX-10 and Rene N6. Generally speaking, the modern alloys are characterised by significantly lower concentrations of Cr and higher concentrations of Al and Re. Concentrations of Ti and Mo are now at very modest levels. The period since 2000 has seen the emergence of the fourth-generation single-crystal superalloys, such as MC-NG, EPM-102 and TMS-162, which are characterised by additions of ruthenium.

Table 1.4 lists the chemical compositions of common wrought superalloys, which are employed therefore in polycrystalline form. When their compositions are compared with those of the cast superalloys, various points emerge. The concentrations of the γ' hardening

Table 1.3. *The compositions (in weight%) of some common cast superalloys*

Alloy	Cr	Co	Mo	W	Al	Ti	Ta	Nb	Re	Ru	Hf	C	B	Zr	Ni
AM1	7.0	8.0	2.0	5.0	5.0	1.8	8.0	1.0	—	—	—	—	—	—	Bal
AM3	8.0	5.5	2.25	5.0	6.0	2.0	3.5	—	—	—	—	—	—	—	Bal
CM186LC	6.0	9.3	0.5	8.4	5.7	0.7	3.4	—	3.0	—	1.4	0.07	0.015	0.005	Bal
CM247LC	8.0	9.3	0.5	9.5	5.6	0.7	3.2	—	—	—	1.4	0.07	0.015	0.010	Bal
CMSX-2	8.0	5.0	0.6	8.0	5.6	1.0	6.0	—	—	—	—	—	—	—	Bal
CMSX-3	8.0	4.8	0.6	8.0	5.6	1.0	6.3	—	—	—	0.1	—	—	—	Bal
CMSX-4	6.5	9.6	0.6	6.4	5.6	1.0	6.5	—	3.0	—	0.1	—	—	—	Bal
CMSX-6	10.0	5.0	3.0	—	4.8	4.7	6.0	—	—	—	0.1	—	—	—	Bal
CMSX-10	2.0	3.0	0.4	5.0	5.7	0.2	8.0	—	6.0	—	0.03	—	—	—	Bal
EPM-102	2.0	16.5	2.0	6.0	5.55	—	8.25	—	5.95	3.0	0.15	0.03	—	—	Bal
GTD-111	14.0	9.5	1.5	3.8	3.0	5.0	3.15	0.07	—	—	—	0.10	0.014	0.007	Bal
GTD-222	22.5	19.1	—	2.0	1.2	2.3	0.94	0.8	—	—	—	0.08	0.004	0.02	Bal
IN100	10.0	15.0	3.0	—	5.5	4.7	—	—	—	—	—	0.18	0.014	0.06	Bal
IN-713LC	12.0	—	4.5	—	5.9	0.6	—	2.0	—	—	—	0.05	0.01	0.10	Bal
IN-738LC	16.0	8.5	1.75	2.6	3.4	3.4	1.75	0.9	—	—	—	0.11	0.01	0.04	Bal
IN-792	12.4	9.2	1.9	3.9	3.5	3.9	4.2	—	—	—	—	0.07	0.016	0.018	Bal
IN-939	22.4	19.0	—	2.0	1.9	3.7	—	1.0	—	—	—	0.15	0.009	0.10	Bal
Mar-M002	8.0	10.0	—	10.0	5.5	1.5	2.6	—	—	—	1.5	0.15	0.015	0.03	Bal
Mar-M246	9.0	10.0	2.5	10.0	5.5	1.5	1.5	—	—	—	1.5	0.15	0.015	0.05	Bal
Mar-M247	8.0	10.0	0.6	10.0	5.5	1.0	3.0	—	—	—	1.5	0.15	0.015	0.03	Bal
Mar-M200Hf	8.0	9.0	—	12.0	5.0	1.9	—	1.0	—	—	2.0	0.13	0.015	0.03	Bal
Mar-M421	15.0	10.8	1.8	3.3	4.5	1.6	—	2.3	—	—	—	0.18	0.019	0.04	Bal
MC2	8.0	5.0	2.0	8.0	5.0	1.5	6.0	—	—	—	0.1	—	—	—	Bal
MC-NG	4.0	—	1.0	5.0	6.0	0.5	5.0	—	4.0	4.0	0.1	—	—	—	Bal
MX4	2.0	16.5	2.0	6.0	5.55	—	8.25	—	5.95	3.0	0.15	0.03	—	—	Bal

Table 1.3. (*cont.*)

Alloy	Cr	Co	Mo	W	Al	Ti	Ta	Nb	Re	Ru	Hf	C	B	Zr	Ni
Nasair 100	9.0	—	1.0	10.5	5.75	1.2	3.3	—	—	—	—	—	—	—	Bal
PWA1422	9.0	10.0	—	12.0	5.0	2.0	—	1.0	—	—	1.5	0.14	0.015	0.1	Bal
PWA1426	6.5	10.0	1.7	6.5	6.0	—	4.0	—	3.0	—	1.5	0.10	0.015	0.1	Bal
PWA1480	10.0	5.0	—	4.0	5.0	1.5	12.0	—	—	—	—	—	—	—	Bal
PWA1483	12.2	9.2	1.9	3.8	3.6	4.2	5.0	—	—	—	—	0.07	—	—	Bal
PWA1484	5.0	10.0	2.0	6.0	5.6	—	9.0	—	3.0	—	0.1	—	—	—	Bal
PWA1487	5.0	10.0	1.9	5.9	5.6	—	8.4	—	3.0	—	0.25	—	—	—	Bal
PWA1497	2.0	16.5	2.0	6.0	5.55	—	8.25	—	5.95	3.0	0.15	0.03	—	—	Bal
Rene 80	14.0	9.0	4.0	4.0	3.0	4.7	—	—	—	—	0.8	0.16	0.015	0.01	Bal
Rene 125	9.0	10.0	2.0	7.0	1.4	2.5	3.8	—	—	—	0.05	0.11	0.017	0.05	Bal
Rene 142	6.8	12.0	1.5	4.9	6.15	—	6.35	—	2.8	—	1.5	0.12	0.015	0.02	Bal
Rene 220	18.0	12.0	3.0	—	0.5	1.0	3.0	5.0	—	—	—	0.02	0.010	—	Bal
Rene N4	9.0	8.0	2.0	6.0	3.7	4.2	4.0	0.5	—	—	—	—	—	—	Bal
Rene N5	7.0	8.0	2.0	5.0	6.2	—	7.0	—	3.0	—	0.2	—	—	—	Bal
Rene N6	4.2	12.5	1.4	6.0	5.75	—	7.2	—	5.4	—	0.15	0.05	0.004	—	Bal
RR2000	10.0	15.0	3.0	—	5.5	4.0	—	—	—	—	—	—	—	—	Bal
SRR99	8.0	5.0	—	10.0	5.5	2.2	12.0	—	—	—	—	—	—	—	Bal
TMS-75	3.0	12.0	2.0	6.0	6.0	—	6.0	—	5.0	—	0.1	—	—	—	Bal
TMS-138	2.9	5.9	2.9	5.9	5.9	—	5.6	—	4.9	2.0	0.1	—	—	—	Bal
TMS-162	2.9	5.8	3.9	5.8	5.8	—	5.6	—	4.9	6.0	0.09	—	—	—	Bal